Names: Yousef Rawashdeh, Maliha Sara, Mandy Wang
NetIDs: mw5288, yhr2004, ms15380

1) Database server:
   Our database server is hosted locally with mySQL and managed with phpMyAdmin. It will be via a website.

```
DESC announcement;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
AnnouncementID	char(5)	NO	PRI	*NULL*	
AnnouncementTitle	varchar(30)	YES		*NULL*	
AnnouncementBody	text	YES		*NULL*	
AnnouncementDate	date	YES		*NULL*	
UserID	char(5)	YES	MUL	*NULL*	
ClubID	char(5)	YES	MUL	*NULL*	

←T→			AnnouncementID	AnnouncementTitle	AnnouncementBody	AnnouncementDate	UserID	ClubID
Edit	Copy	Delete	AN001	Welcome!	We are excited to welcome new members to our club!...	2023-08-25	U0001	CL001
Edit	Copy	Delete	AN002	Upcoming Event: Networking Nig	Join us for a night of networking with industry pr...	2023-09-10	U0001	CL001
Edit	Copy	Delete	AN003	Volunteer Opportunity	We are looking for volunteers to help with our com...	2023-09-20	U0002	CL009
Edit	Copy	Delete	AN004	Study & Chill Session	Come hang out before finals!	2023-12-01	U0003	CL001
Edit	Copy	Delete	AN005	Guest Speaker Event	We are excited to announce that we will be hosting...	2023-10-15	U0001	CL009
Edit	Copy	Delete	AN006	Hackathon	Get ready for our hackathon! More details to come ...	2023-11-01	U0004	CL007
Edit	Copy	Delete	AN007	Volunteer Opportunity	Join us for a day of mentoring local high school s...	2023-10-05	U0002	CL009
Edit	Copy	Delete	AN008	End of Semester Party	Join us for our end of semester party! Food and dr...	2023-12-15	U0001	CL001
Edit	Copy	Delete	AN009	Resume Workshop	Join us for a resume workshop to help you prepare ...	2023-09-30	U0002	CL002
Edit	Copy	Delete	AN010	Arduino Workshop	Learn the basics of Arduino programming and build ...	2023-10-20	U0002	CL008

```
DESC attendance;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
CheckInTime	timestamp	NO		current_timestamp()	on update current_timestamp()
CheckInMethod	varchar(20)	YES		*NULL*	
EventID	char(5)	NO	PRI	*NULL*	
RSVPID	char(5)	NO	PRI	*NULL*	

←T→				CheckInTime	CheckInMethod	EventID	RSVPID
☐	Edit	Copy	Delete	2025-10-15 17:55:00	Manual	EV001	RS001
☐	Edit	Copy	Delete	2025-10-15 18:02:00	Manual	EV001	RS002
☐	Edit	Copy	Delete	2025-10-20 16:58:00	QRCode	EV002	RS004
☐	Edit	Copy	Delete	2025-11-05 13:50:00	Manual	EV003	RS005
☐	Edit	Copy	Delete	2025-09-28 10:55:00	QRCode	EV004	RS006
☐	Edit	Copy	Delete	2025-10-08 15:58:00	Manual	EV005	RS007
☐	Edit	Copy	Delete	2025-11-12 18:55:00	QRCode	EV006	RS009
☐	Edit	Copy	Delete	2025-10-22 17:50:00	Manual	EV007	RS010
☐	Edit	Copy	Delete	2025-11-01 11:50:00	Manual	EV008	RS011
☐	Edit	Copy	Delete	2025-09-18 14:55:00	QRCode	EV009	RS012

```
SELECT COUNT(*) from Category;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [R

Extra options

COUNT(*)
10

←T→				CategoryID	CategoryName
☐	Edit	Copy	Delete	CA001	Sports
☐	Edit	Copy	Delete	CA002	Arts
☐	Edit	Copy	Delete	CA003	Academic
☐	Edit	Copy	Delete	CA004	Cultural
☐	Edit	Copy	Delete	CA005	Community Service
☐	Edit	Copy	Delete	CA006	Professional Develop
☐	Edit	Copy	Delete	CA007	Hobbies
☐	Edit	Copy	Delete	CA008	Media
☐	Edit	Copy	Delete	CA009	Leadership
☐	Edit	Copy	Delete	CA010	STEM

```
DESC club;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
ClubID	char(5)	NO	PRI	*NULL*	
ClubName	varchar(30)	YES		*NULL*	
ClubDescription	text	YES		*NULL*	
ClubCreationDate	date	YES		*NULL*	

←T→	ClubID	ClubName	ClubDescription	ClubCreationDate
Edit Copy Delete	CL001	Society of Asian Scientists &	SASE at NYU Tandon School of Engineering aims to e...	2020-01-15
Edit Copy Delete	CL002	Alpha Omega Epsilon (AOE)	Alpha Omega Epsilon is a professional and social s...	2019-09-10
Edit Copy Delete	CL003	Business & Finance Group (BFG)	The Business & Finance Club at NYU is dedicated to...	2021-02-20
Edit Copy Delete	CL004	Chemists Club	The Chemists Club strives to bridge the gap betwee...	2020-10-05
Edit Copy Delete	CL005	oSTEM	The purpose of oSTEM at NYU is to foster a campus ...	2019-09-10
Edit Copy Delete	CL006	Poly Anime Society (PAS)	The Poly Anime Society promotes an environment tha...	2021-01-25
Edit Copy Delete	CL007	Poly Programming Club (PPC)	The Poly Programming Club (PPC) is the official co...	2022-12-23
Edit Copy Delete	CL008	Robotics Club	Robotics Clubs mission is to create a relaxed lear...	2020-09-15
Edit Copy Delete	CL009	Society of Women Engineers (SW	The Society of Women Engineers (SWE) is a not-for-...	2019-09-10
Edit Copy Delete	CL010	Undergraduate Student Council	The School of Engineering Student Councils are ele...	2018-09-11

```
DESC clubcategory;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
ClubID	char(5)	NO	PRI	*NULL*	
CategoryID	char(5)	NO	PRI	*NULL*	

←T→				ClubID	CategoryID
☐	Edit	Copy	Delete	CL001	CA004
☐	Edit	Copy	Delete	CL001	CA006
☐	Edit	Copy	Delete	CL002	CA006
☐	Edit	Copy	Delete	CL002	CA009
☐	Edit	Copy	Delete	CL003	CA006
☐	Edit	Copy	Delete	CL004	CA010
☐	Edit	Copy	Delete	CL005	CA006
☐	Edit	Copy	Delete	CL005	CA010
☐	Edit	Copy	Delete	CL006	CA007
☐	Edit	Copy	Delete	CL007	CA010
☐	Edit	Copy	Delete	CL008	CA010
☐	Edit	Copy	Delete	CL009	CA006
☐	Edit	Copy	Delete	CL010	CA009

```
DESC clubmembership;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
UserID	char(5)	NO	PRI	*NULL*	
ClubID	char(5)	NO	PRI	*NULL*	
MembershipRole	varchar(20)	YES		*NULL*	
MembershipStatus	varchar(10)	YES		*NULL*	
MembershipJoinDate	date	YES		*NULL*	

```
SELECT * FROM `clubmembership`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

←T→				UserID	ClubID	MembershipRole	MembershipStatus	MembershipJoinDate
Edit	Copy	Delete		U0001	CL001	Owner	Active	2026-04-01
Edit	Copy	Delete		U0002	CL001	Officer	Active	2026-04-01
Edit	Copy	Delete		U0003	CL001	Member	Active	2026-04-02
Edit	Copy	Delete		U0004	CL001	Member	Active	2026-04-02
Edit	Copy	Delete		U0005	CL002	Owner	Active	2026-04-01
Edit	Copy	Delete		U0006	CL002	Officer	Active	2026-04-02
Edit	Copy	Delete		U0007	CL002	Member	Active	2026-04-03
Edit	Copy	Delete		U0008	CL003	Owner	Active	2026-04-01
Edit	Copy	Delete		U0009	CL003	Member	Active	2026-04-03
Edit	Copy	Delete		U0010	CL003	Member	Inactive	2026-04-04

```
DESC event;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
EventID	char(5)	NO	PRI	*NULL*	
EventTitle	varchar(30)	YES		*NULL*	
EventDescription	text	YES		*NULL*	
EventStartTime	timestamp	NO		current_timestamp()	on update current_timestamp()
EventEndTime	timestamp	NO		0000-00-00 00:00:00	
EventCapacity	int(11)	YES		*NULL*	
EventStatus	varchar(10)	YES		*NULL*	
EventVisibility	varchar(10)	YES		*NULL*	
EventTypeID	char(5)	YES	MUL	*NULL*	
LocationID	char(5)	YES	MUL	*NULL*	
ClubID	char(5)	YES	MUL	*NULL*	

←T→				EventID	EventTitle	EventDescription	EventStartTime	EventEndTime	EventCapacity	EventStatus	EventVisibility	EventTypeID	LocationID	ClubID
☐	Edit	Copy	Delete	EV001	SASE Networking Night	Connect with industry professionals and SASE alumn...	2025-10-15 18:00:00	2025-10-15 20:00:00	80	Scheduled	Public	ET002	LO005	CL001
☐	Edit	Copy	Delete	EV002	AOE Sisters Social	A members-only social for current AOE sisters.	2025-10-20 17:00:00	2025-10-20 19:00:00	30	Scheduled	MembersOnl	ET003	LO002	CL002
☐	Edit	Copy	Delete	EV003	BFG Stock Pitch Competition	Compete to deliver the best stock pitch in front o...	2025-11-05 14:00:00	2025-11-05 17:00:00	50	Scheduled	Public	ET007	LO005	CL003
☐	Edit	Copy	Delete	EV004	Chem Club Lab Tour	Tour the chemistry research labs with faculty guid...	2025-09-28 11:00:00	2025-09-28 12:30:00	20	Completed	MembersOnl	ET009	LO007	CL004
☐	Edit	Copy	Delete	EV005	oSTEM Study Session	Midterm study session open to all students. Snacks...	2025-10-08 16:00:00	2025-10-08 19:00:00	40	Completed	Public	ET003	LO003	CL005
☐	Edit	Copy	Delete	EV006	Anime Movie Night	Watch a classic Miyazaki film with club members. P...	2025-11-12 19:00:00	2025-11-12 22:00:00	60	Scheduled	Public	ET003	LO005	CL006
☐	Edit	Copy	Delete	EV007	PPC Weekly Practice	Weekly competitive programming practice session. A...	2025-10-22 18:00:00	2025-10-22 20:00:00	35	Scheduled	MembersOnl	ET004	LO008	CL007
☐	Edit	Copy	Delete	EV008	Robotics Arduino Workshop	Hands-on Arduino workshop. Build your first robot ...	2025-11-01 13:00:00	2025-11-01 16:00:00	25	Scheduled	Public	ET001	LO004	CL008
☐	Edit	Copy	Delete	EV009	SWE Resume Workshop	Get your resume reviewed by engineers from top com...	2025-09-18 15:00:00	2025-09-18 17:00:00	45	Completed	Public	ET001	LO002	CL009
☐	Edit	Copy	Delete	EV010	Student Council Town Hall	Open forum for students to voice concerns and idea...	2025-10-01 12:00:00	2025-10-01 13:30:00	100	Completed	Public	ET004	LO005	CL010
☐	Edit	Copy	Delete	EV011	SASE Hackathon 2025	A 24-hour hackathon hosted by SASE. Teams of up to...	2025-11-15 10:00:00	2025-11-16 10:00:00	120	Scheduled	Public	ET005	LO006	CL001
☐	Edit	Copy	Delete	EV012	SWE Industry Panel	Hear from women engineers at Meta, Google, and Ama...	2025-11-20 17:00:00	2025-11-20 19:00:00	70	Scheduled	Public	ET006	LO005	CL009

```
DESC eventtype;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
EventTypeID	char(5)	NO	PRI	*NULL*	
EventTypeName	varchar(20)	YES		*NULL*	

```
SELECT * FROM `eventtype`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refr

Show all | Number of rows: 25 Filter rows: Search

Extra options

←T→				EventTypeID	EventTypeName
	Edit	Copy	Delete	ET001	Workshop
	Edit	Copy	Delete	ET002	Networking
	Edit	Copy	Delete	ET003	Social
	Edit	Copy	Delete	ET004	General Meeting
	Edit	Copy	Delete	ET005	Hackathon
	Edit	Copy	Delete	ET006	Seminar
	Edit	Copy	Delete	ET007	Competition
	Edit	Copy	Delete	ET008	Community Service
	Edit	Copy	Delete	ET009	Info Session
	Edit	Copy	Delete	ET010	Conference

```
DESC joinrequest;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
RequestID	char(5)	NO	PRI	*NULL*	
RequestStatus	varchar(10)	YES		*NULL*	
RequestTime	timestamp	NO		current_timestamp()	on update current_timestamp()
UserID	char(5)	YES	MUL	*NULL*	
ClubID	char(5)	YES	MUL	*NULL*	

```
SELECT * FROM `joinrequest`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key:

Extra options

←T→	RequestID	RequestStatus	RequestTime	UserID	ClubID
Edit Copy Delete	R0001	Pending	2026-04-06 10:00:00	U0004	CL002
Edit Copy Delete	R0002	Approved	2026-04-06 11:00:00	U0003	CL003
Edit Copy Delete	R0003	Rejected	2026-04-06 12:00:00	U0007	CL001
Edit Copy Delete	R0004	Pending	2026-04-07 09:30:00	U0009	CL001
Edit Copy Delete	R0005	Approved	2026-04-07 10:15:00	U0010	CL002
Edit Copy Delete	R0006	Pending	2026-04-07 11:45:00	U0006	CL003
Edit Copy Delete	R0007	Rejected	2026-04-08 01:00:00	U0002	CL003
Edit Copy Delete	R0008	Pending	2026-04-08 02:20:00	U0005	CL001
Edit Copy Delete	R0009	Approved	2026-04-08 03:10:00	U0008	CL002
Edit Copy Delete	R0010	Pending	2026-04-08 04:00:00	U0001	CL003

```
SELECT COUNT(*) from Attendance;
```

☐ Profiling [Edit inline] [Edit] [Explain S

Extra options

COUNT(*)
10

```
DESC location;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
LocationID	char(5)	NO	PRI	*NULL*	
BuildingName	varchar(30)	YES		*NULL*	
RoomNumber	int(11)	YES		*NULL*	
HomeAddress	varchar(50)	YES		*NULL*	

```
SELECT COUNT(*) from Location;
```

Profiling [Edit inline] [Edit] [Explain SQ

Extra options

COUNT(*)
10

```
SELECT * FROM `location`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

←T→	LocationID	BuildingName	RoomNumber	HomeAddress
Edit Copy Delete	LO001	Rogers Hall	101	6 MetroTech Center, Brooklyn, NY 11201
Edit Copy Delete	LO002	Jacobs Academic Building	674	370 Jay Street, Brooklyn, NY 11201
Edit Copy Delete	LO003	Dibner Library	201	5 MetroTech Center, Brooklyn, NY 11201
Edit Copy Delete	LO004	Tandon MakerSpace	100	6 MetroTech Center, Brooklyn, NY 11201
Edit Copy Delete	LO005	Pfizer Auditorium	1	5 MetroTech Center, Brooklyn, NY 11201
Edit Copy Delete	LO006	2 MetroTech Center	802	2 MetroTech Center, Brooklyn, NY 11201
Edit Copy Delete	LO007	Rogers Hall	310	6 MetroTech Center, Brooklyn, NY 11201
Edit Copy Delete	LO008	Jacobs Academic Building	473	370 Jay Street, Brooklyn, NY 11201
Edit Copy Delete	LO009	Dibner Library	105	5 MetroTech Center, Brooklyn, NY 11201
Edit Copy Delete	LO010	2 MetroTech Center	1001	2 MetroTech Center, Brooklyn, NY 11201

```
SELECT COUNT(*) from User;
```

☐ Profiling [Edit inline] [Edit] [E

Extra options

COUNT(*)
10

```
DESC rsvp;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
RSVPID	char(5)	NO	PRI	*NULL*	
RSVPStatus	varchar(10)	YES		*NULL*	
RSVPCreationDate	date	YES		*NULL*	
UserID	char(5)	YES	MUL	*NULL*	
EventID	char(5)	YES	MUL	*NULL*	

```
SELECT COUNT(*) from RSVP;
```

Profiling [Edit inline] [Edit] [Explain SQ

Extra options

COUNT(*)
12

```
SELECT * FROM `rsvp`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key:

Extra options

←T→			RSVPID	RSVPStatus	RSVPCreationDate	UserID	EventID
Edit	Copy	Delete	RS001	Going	2025-10-10	U0001	EV001
Edit	Copy	Delete	RS002	Going	2025-10-11	U0002	EV001
Edit	Copy	Delete	RS003	Tentative	2025-10-12	U0003	EV001
Edit	Copy	Delete	RS004	Going	2025-10-18	U0002	EV002
Edit	Copy	Delete	RS005	Going	2025-10-30	U0004	EV003
Edit	Copy	Delete	RS006	Going	2025-09-25	U0003	EV004
Edit	Copy	Delete	RS007	Going	2025-10-05	U0005	EV005
Edit	Copy	Delete	RS008	NotGoing	2025-10-06	U0006	EV005
Edit	Copy	Delete	RS009	Going	2025-11-10	U0007	EV006
Edit	Copy	Delete	RS010	Going	2025-10-20	U0008	EV007
Edit	Copy	Delete	RS011	Tentative	2025-10-28	U0009	EV008
Edit	Copy	Delete	RS012	Going	2025-09-15	U0010	EV009

```
DESC user;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
UserID	char(5)	NO	PRI	NULL	
FirstName	varchar(20)	YES		NULL	
LastName	varchar(20)	YES		NULL	
Email	varchar(20)	YES		NULL	
PasswordHash	char(20)	YES		NULL	
AccountCreationDate	date	YES		NULL	

```
SELECT * FROM `user`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

			UserID	FirstName	LastName	Email	PasswordHash	AccountCreationDate
Edit	Copy	Delete	U0001	Yousef	Rawashdeh	yousef@nyu.edu	hashed_pw_1	2026-04-01
Edit	Copy	Delete	U0002	Mandy	Wang	mandy@nyu.edu	hashed_pw_2	2026-04-01
Edit	Copy	Delete	U0003	Maliha	Khan	maliha@nyu.edu	hashed_pw_3	2026-04-02
Edit	Copy	Delete	U0004	Adam	Lee	adam@nyu.edu	hashed_pw_4	2026-04-02
Edit	Copy	Delete	U0005	Sara	Ali	sara@nyu.edu	hashed_pw_5	2026-04-03
Edit	Copy	Delete	U0006	Omar	Hassan	omar@nyu.edu	hashed_pw_6	2026-04-03
Edit	Copy	Delete	U0007	Lina	Park	lina@nyu.edu	hashed_pw_7	2026-04-04
Edit	Copy	Delete	U0008	Noah	Smith	noah@nyu.edu	hashed_pw_8	2026-04-04
Edit	Copy	Delete	U0009	Aisha	Patel	aisha@nyu.edu	hashed_pw_9	2026-04-05
Edit	Copy	Delete	U0010	Daniel	Kim	daniel@nyu.edu	hashed_pw_10	2026-04-05

```
SELECT COUNT(*) from Club;
```

Profiling [Edit inline] [Edit] [

Extra options

COUNT(*)
10

```
SELECT COUNT(*) from ClubMembership;
```

Profiling [Edit inline] [Edit] [Explain SQL] [

Extra options

COUNT(*)
10

```
SELECT COUNT(*) from Announcement;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Cr

Extra options

COUNT(*)
10

```
SELECT COUNT(*) from EventType;
```

☐ Profiling [Edit inline] [Edit] [Explain

Extra options

COUNT(*)
10

```
SELECT COUNT(*) from Event;
```

☐ Profiling [Edit inline] [Edit] [Explain S

Extra options

COUNT(*)
12

```
SELECT COUNT(*) from ClubCategory;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [R

Extra options

COUNT(*)
13

```
SELECT COUNT(*) from JoinRequest;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Cre

Extra options

COUNT(*)
10

Types of Commands Used:

- SELECT
- INSERT INTO
- UPDATE, SET, WHERE
- CREATE TRIGGER
- STORED PROCEDURE
- CREATE TABLE
- JOIN commands
-

<u>Database Table Creation:</u>

```
CREATE DATABASE club_organizations;
USE club_organizations;
CREATE TABLE User (
   UserID CHAR(5),
   FirstName VARCHAR(20),
   LastName VARCHAR(20),
   Email VARCHAR(20),
   PasswordHash CHAR(20),
   AccountCreationDate DATE,
   Primary Key (UserID)
);
```

```
CREATE TABLE Category (
   CategoryID CHAR(5),
   CategoryName VARCHAR(20),
   Primary Key (CategoryID)
);
CREATE TABLE Club (
   ClubID CHAR(5),
   ClubName VARCHAR(30),
   ClubDescription TEXT,
   ClubCreationDate DATE,
   Primary Key (ClubID)
);

CREATE TABLE ClubCategory (
   ClubID CHAR(5),
   CategoryID CHAR(5),
   Primary Key (ClubID, CategoryID),
   FOREIGN KEY (ClubID) REFERENCES Club(ClubID),
   FOREIGN KEY (CategoryID) REFERENCES Category(CategoryID)
);

CREATE TABLE JoinRequest (
   RequestID CHAR(5),
   RequestStatus VARCHAR(10),
   RequestTime TIMESTAMP,
   UserID CHAR(5),
   ClubID CHAR(5),
   Primary Key (RequestID),
   FOREIGN KEY (UserID) REFERENCES User(UserID),
   FOREIGN KEY (ClubID) REFERENCES Club(ClubID)
);
CREATE TABLE ClubMembership (
   UserID CHAR(5),
   ClubID CHAR(5),
   MembershipRole VARCHAR(20),
   MembershipStatus VARCHAR(10),
   MembershipJoinDate DATE,
   Primary Key (UserID, ClubID),
   FOREIGN KEY (UserID) REFERENCES User(UserID),
   FOREIGN KEY (ClubID) REFERENCES Club(ClubID)
```

```
);
CREATE TABLE Announcement (
    AnnouncementID CHAR(5),
    AnnouncementTitle VARCHAR(30),
    AnnouncementBody TEXT,
    AnnouncementDate DATE,
    UserID CHAR(5),
    ClubID CHAR(5),
    Primary Key (AnnouncementID),
    FOREIGN KEY (UserID) REFERENCES User(UserID),
    FOREIGN KEY (ClubID) REFERENCES Club(ClubID)
);
CREATE TABLE Location (
    LocationID CHAR(5),
    BuildingName VARCHAR(30),
    RoomNumber INT,
    HomeAddress VARCHAR(50),
    Primary Key (LocationID)
);
CREATE TABLE EventType (
    EventTypeID CHAR(5),
    EventTypeName VARCHAR(20),
    Primary Key (EventTypeID)
);
CREATE TABLE Event (
    EventID CHAR(5),
    EventTitle VARCHAR(30),
    EventDescription TEXT,
    EventStartTime TIMESTAMP,
    EventEndTime TIMESTAMP,
    EventCapacity INT,
    EventStatus VARCHAR(10),
    EventVisibility VARCHAR(10),
    EventTypeID CHAR(5),
    LocationID CHAR(5),
    ClubID CHAR(5),
    Primary Key (EventID),
    FOREIGN KEY (EventTypeID) REFERENCES EventType(EventTypeID),
    FOREIGN KEY (LocationID) REFERENCES Location(LocationID),
    FOREIGN KEY (ClubID) REFERENCES Club(ClubID)
```

```
);
CREATE TABLE RSVP (
   RSVPID CHAR(5),
   RSVPStatus VARCHAR(10),
   RSVPCreationDate DATE,
   UserID CHAR(5),
   EventID CHAR(5),
   Primary Key (RSVPID),
   UNIQUE (RSVPID, EventID),
   FOREIGN KEY (UserID) REFERENCES User(UserID),
   FOREIGN KEY (EventID) REFERENCES Event(EventID)
);
CREATE TABLE Attendance (
   CheckInTime TIMESTAMP,
   CheckInMethod VARCHAR(20),
   EventID CHAR(5),
   RSVPID CHAR(5),
   PRIMARY KEY (EventID, RSVPID),
   FOREIGN KEY (RSVPID, EventID) REFERENCES RSVP(RSVPID, EventID)
);
```

Basic SQL commands used for Clubs, Announcements, and Club Categories:

```
-- CLUB CATEGORIES
INSERT INTO Category (CategoryID, CategoryName) VALUES ('CA001', 'Sports');
INSERT INTO Category (CategoryID, CategoryName) VALUES ('CA002', 'Arts');
INSERT INTO Category (CategoryID, CategoryName) VALUES ('CA003', 'Academic');
INSERT INTO Category (CategoryID, CategoryName) VALUES ('CA004', 'Cultural');
INSERT INTO Category (CategoryID, CategoryName) VALUES ('CA005', 'Community
Service');
INSERT INTO Category (CategoryID, CategoryName) VALUES ('CA006', 'Professional
Development');
INSERT INTO Category (CategoryID, CategoryName) VALUES ('CA007', 'Hobbies');
INSERT INTO Category (CategoryID, CategoryName) VALUES ('CA008', 'Media');
INSERT INTO Category (CategoryID, CategoryName) VALUES ('CA009', 'Leadership');
INSERT INTO Category (CategoryID, CategoryName) VALUES ('CA010', 'STEM');

-- CLUBS AND ASSOCIATED CATEGORIES
INSERT INTO Club (ClubID, ClubName, ClubDescription, ClubCreationDate) VALUES
('CL001', 'Society of Asian Scientists & Engineers (SASE)',
```

'SASE at NYU Tandon School of Engineering aims to equip members with the skill set to succeed in professional environments',
'2020-01-15');
INSERT INTO ClubCategory (ClubID, CategoryID) VALUES ('CL001', 'CA004');
INSERT INTO ClubCategory (ClubID, CategoryID) VALUES ('CL001', 'CA006');

INSERT INTO Club (ClubID, ClubName, ClubDescription, ClubCreationDate) VALUES ('CL002', 'Alpha Omega Epsilon (AOE)', 'Alpha Omega Epsilon is a professional and social sorority', '2019-09-10');
INSERT INTO ClubCategory (ClubID, CategoryID) VALUES ('CL002', 'CA006');
INSERT INTO ClubCategory (ClubID, CategoryID) VALUES ('CL002', 'CA009');

INSERT INTO Club (ClubID, ClubName, ClubDescription, ClubCreationDate) VALUES ('CL003', 'Business & Finance Group (BFG)',
'The Business & Finance Club at NYU is dedicated to empowering the NYU Tandon student body with comprehensive financial education',
'2021-02-20');
INSERT INTO ClubCategory (ClubID, CategoryID) VALUES ('CL003', 'CA006');

INSERT INTO Club (ClubID, ClubName, ClubDescription, ClubCreationDate) VALUES ('CL004', 'Chemists Club',
'The Chemists Club strives to bridge the gap between academia and industry in the chemistry and chemical engineering field',
'2020-10-05');
INSERT INTO ClubCategory (ClubID, CategoryID) VALUES ('CL004', 'CA010');

INSERT INTO Club (ClubID, ClubName, ClubDescription, ClubCreationDate) VALUES ('CL005', 'oSTEM',
'The purpose of oSTEM at NYU is to foster a campus and community environment that is safe, supportive,
and informative to all students regardless of physical gender, perceived gender, sexual orientation,
or any other physical or physiological attribute', '2019-09-10');
INSERT INTO ClubCategory (ClubID, CategoryID) VALUES ('CL005', 'CA006');
INSERT INTO ClubCategory (ClubID, CategoryID) VALUES ('CL005', 'CA010');

INSERT INTO Club (ClubID, ClubName, ClubDescription, ClubCreationDate) VALUES ('CL006', 'Poly Anime Society (PAS)',
'The Poly Anime Society promotes an environment that heightens awareness and appreciation of Japanese culture and entertainment',

'2021-01-25');
INSERT INTO ClubCategory (ClubID, CategoryID) VALUES ('CL006', 'CA007');

INSERT INTO Club (ClubID, ClubName, ClubDescription, ClubCreationDate) VALUES
('CL007', 'Poly Programming Club (PPC)',
'The Poly Programming Club (PPC) is the official competitive programming club of NYU Tandon!',
'2022-12-23');
INSERT INTO ClubCategory (ClubID, CategoryID) VALUES ('CL007', 'CA010');

INSERT INTO Club (ClubID, ClubName, ClubDescription, ClubCreationDate) VALUES
('CL008', 'Robotics Club',
'Robotics Clubs mission is to create a relaxed learning environment that encourages students to learn from one another.',
'2020-09-15');
INSERT INTO ClubCategory (ClubID, CategoryID) VALUES ('CL008', 'CA010');

INSERT INTO Club (ClubID, ClubName, ClubDescription, ClubCreationDate) VALUES
('CL009', 'Society of Women Engineers (SWE)',
'The Society of Women Engineers (SWE) is a not-for-profit educational and service organization that
empowers women to succeed and advance in the field of engineering', '2019-09-10');
INSERT INTO ClubCategory (ClubID, CategoryID) VALUES ('CL009', 'CA006');

INSERT INTO Club (ClubID, ClubName, ClubDescription, ClubCreationDate) VALUES
('CL010', 'Undergraduate Student Council',
'The School of Engineering Student Councils are elected by students of the School of Engineering to serve as
 their primary representatives, advocates, and liaisons to the entire student community',
'2018-09-11');
INSERT INTO ClubCategory (ClubID, CategoryID) VALUES ('CL010', 'CA009');

-- ANNOUNCEMENTS
INSERT INTO Announcement (AnnouncementID, AnnouncementTitle, AnnouncementBody, AnnouncementDate, UserID, ClubID) VALUES
('AN001', 'Welcome!', 'We are excited to welcome new members to our club! Join us for our first meeting of the semester',
'2023-08-25', 'U0001', 'CL001');

INSERT INTO Announcement (AnnouncementID, AnnouncementTitle, AnnouncementBody, AnnouncementDate, UserID, ClubID) VALUES
('AN002', 'Upcoming Event: Networking Night', 'Join us for a night of networking with industry professionals!',
'2023-09-10', 'U0001', 'CL001');

INSERT INTO Announcement (AnnouncementID, AnnouncementTitle, AnnouncementBody, AnnouncementDate, UserID, ClubID) VALUES
('AN003', 'Volunteer Opportunity', 'We are looking for volunteers to help with our community service event next month.
Sign up if you are interested!', '2023-09-20', 'U0002', 'CL009');

INSERT INTO Announcement (AnnouncementID, AnnouncementTitle, AnnouncementBody, AnnouncementDate, UserID, ClubID) VALUES
('AN004', 'Study & Chill Session', 'Come hang out before finals!', '2023-12-01', 'U0003', 'CL001');

INSERT INTO Announcement (AnnouncementID, AnnouncementTitle, AnnouncementBody, AnnouncementDate, UserID, ClubID) VALUES
('AN005', 'Guest Speaker Event', 'We are excited to announce that we will be hosting a guest speaker from Google next week!',
'2023-10-15', 'U0001', 'CL009');

INSERT INTO Announcement (AnnouncementID, AnnouncementTitle, AnnouncementBody, AnnouncementDate, UserID, ClubID) VALUES
('AN006', 'Hackathon', 'Get ready for our hackathon! More details to come soon.', '2023-11-01', 'U0004', 'CL007');

INSERT INTO Announcement (AnnouncementID, AnnouncementTitle, AnnouncementBody, AnnouncementDate, UserID, ClubID) VALUES
('AN007', 'Volunteer Opportunity', 'Join us for a day of mentoring local high school students', '2023-10-05', 'U0002', 'CL009');

INSERT INTO Announcement (AnnouncementID, AnnouncementTitle, AnnouncementBody, AnnouncementDate, UserID, ClubID) VALUES
('AN008', 'End of Semester Party', 'Join us for our end of semester party! Food and drinks will be provided.', '2023-12-15', 'U0001', 'CL001');

INSERT INTO Announcement (AnnouncementID, AnnouncementTitle, AnnouncementBody, AnnouncementDate, UserID, ClubID) VALUES

('AN009', 'Resume Workshop', 'Join us for a resume workshop to help you prepare for your job search!', '2023-09-30', 'U0002', 'CL002');

INSERT INTO Announcement (AnnouncementID, AnnouncementTitle, AnnouncementBody, AnnouncementDate, UserID, ClubID) VALUES
('AN010', 'Arduino Workshop', 'Learn the basics of Arduino programming and build your first project!', '2023-10-20', 'U0002', 'CL008');

<u>Advanced SQL commands used for Clubs, Announcements, and Club Categories:</u>

```
-- EDITING CLUB INFORMATION

-- update club name and description for a club
DELIMITER //
CREATE PROCEDURE UpdateClubInfo (
   IN club_id CHAR(5),
   IN club_name VARCHAR(30),
   IN club_description TEXT
)
BEGIN
   IF club_name IS NOT NULL THEN
      UPDATE Club
      SET ClubName = club_name
      WHERE ClubID = club_id;
   END IF;
   IF club_description IS NOT NULL THEN
      UPDATE Club
      SET ClubDescription = club_description
      WHERE ClubID = club_id;
   END IF;
END //

DELIMITER ;

-- search for clubs by name
DELIMITER //
CREATE PROCEDURE ClubSearch (
   IN keyword VARCHAR(50)
)
BEGIN
```

```
    SELECT ClubID, ClubName, ClubDescription
    FROM Club
    WHERE ClubName LIKE keyword
    ORDER BY ClubName;
END //

DELIMITER ;

-- filter clubs by category
DELIMITER //
CREATE PROCEDURE FilterClubsByCategory (
    IN category_name VARCHAR(20)
)
BEGIN
    SELECT c.ClubName, c.ClubDescription
    FROM Club c
    INNER JOIN ClubCategory cc ON c.ClubID = cc.ClubID
    INNER JOIN Category ca ON cc.CategoryID = ca.CategoryID
    WHERE ca.CategoryName = category_name
    ORDER BY c.ClubName;
END //

DELIMITER ;

-- update announcement title and body for an announcement
DELIMITER //
CREATE PROCEDURE UpdateAnnouncementInfo (
    IN announcement_id CHAR(5),
    IN announcement_title VARCHAR(30),
    IN announcement_body TEXT
)
BEGIN
    IF announcement_title IS NOT NULL THEN
        UPDATE Announcement
        SET AnnouncementTitle = announcement_title
        WHERE AnnouncementID = announcement_id;
    END IF;
    IF announcement_body IS NOT NULL THEN
        UPDATE Announcement
        SET AnnouncementBody = announcement_body
```

```
        WHERE AnnouncementID = announcement_id;
    END IF;
END //

DELIMITER ;

-- remove announcement
DELIMITER //
CREATE PROCEDURE RemoveAnnouncement (
    IN announcement_id CHAR(5)
)
BEGIN
    DELETE FROM Announcement
    WHERE AnnouncementID = announcement_id;
END //

DELIMITER ;

-- delete club
DELIMITER //

CREATE PROCEDURE DeleteClub (
    IN club_id CHAR(5)
)
BEGIN
    DELETE FROM ClubMembership
    WHERE ClubID = club_id;

    DELETE FROM ClubCategory
    WHERE ClubID = club_id;

    DELETE FROM Announcement
    WHERE ClubID = club_id;

    DELETE FROM Club
    WHERE ClubID = club_id;
END //

DELIMITER ;
```

SQL commands used in USER:

Basic SQL:

- Select
- Insert
- Update
- Delete

Advanced SQL:

- Stored Procedure (ApproveJoinRequest)
- Trigger (PreventDuplicatePendingJoinRequest)

SQL Code:

User.Sql:

```
-- USER DATA

INSERT INTO User (UserID, FirstName, LastName, Email, PasswordHash,
AccountCreationDate)
VALUES
('U0001', 'Yousef', 'Rawashdeh', 'yousef@nyu.edu', 'hashed_pw_1', '2026-04-01'),
('U0002', 'Mandy', 'Wang', 'mandy@nyu.edu', 'hashed_pw_2', '2026-04-01'),
('U0003', 'Maliha', 'Khan', 'maliha@nyu.edu', 'hashed_pw_3', '2026-04-02'),
('U0004', 'Adam', 'Lee', 'adam@nyu.edu', 'hashed_pw_4', '2026-04-02'),
('U0005', 'Sara', 'Ali', 'sara@nyu.edu', 'hashed_pw_5', '2026-04-03'),
('U0006', 'Omar', 'Hassan', 'omar@nyu.edu', 'hashed_pw_6', '2026-04-03'),
('U0007', 'Lina', 'Park', 'lina@nyu.edu', 'hashed_pw_7', '2026-04-04'),
('U0008', 'Noah', 'Smith', 'noah@nyu.edu', 'hashed_pw_8', '2026-04-04'),
('U0009', 'Aisha', 'Patel', 'aisha@nyu.edu', 'hashed_pw_9', '2026-04-05'),
('U0010', 'Daniel', 'Kim', 'daniel@nyu.edu', 'hashed_pw_10', '2026-04-05');

-- CLUB MEMBERSHIP DATA

INSERT INTO ClubMembership (UserID, ClubID, MembershipRole, MembershipStatus,
MembershipJoinDate)
VALUES
('U0001', 'CL001', 'Owner', 'Active', '2026-04-01'),
```

```
('U0002', 'CL001', 'Officer', 'Active', '2026-04-01'),
('U0003', 'CL001', 'Member', 'Active', '2026-04-02'),
('U0004', 'CL001', 'Member', 'Active', '2026-04-02'),
('U0005', 'CL002', 'Owner', 'Active', '2026-04-01'),
('U0006', 'CL002', 'Officer', 'Active', '2026-04-02'),
('U0007', 'CL002', 'Member', 'Active', '2026-04-03'),
('U0008', 'CL003', 'Owner', 'Active', '2026-04-01'),
('U0009', 'CL003', 'Member', 'Active', '2026-04-03'),
('U0010', 'CL003', 'Member', 'Inactive', '2026-04-04');


-- JOIN REQUEST DATA

INSERT INTO JoinRequest (RequestID, RequestStatus, RequestTime, UserID, ClubID)
VALUES
('R0001', 'Pending', '2026-04-06 10:00:00', 'U0004', 'CL002'),
('R0002', 'Approved', '2026-04-06 11:00:00', 'U0003', 'CL003'),
('R0003', 'Rejected', '2026-04-06 12:00:00', 'U0007', 'CL001'),
('R0004', 'Pending', '2026-04-07 09:30:00', 'U0009', 'CL001'),
('R0005', 'Approved', '2026-04-07 10:15:00', 'U0010', 'CL002'),
('R0006', 'Pending', '2026-04-07 11:45:00', 'U0006', 'CL003'),
('R0007', 'Rejected', '2026-04-08 01:00:00', 'U0002', 'CL003'),
('R0008', 'Pending', '2026-04-08 02:20:00', 'U0005', 'CL001'),
('R0009', 'Approved', '2026-04-08 03:10:00', 'U0008', 'CL002'),
('R0010', 'Pending', '2026-04-08 04:00:00', 'U0001', 'CL003');
-- USER LOGIN

-- Retrieve a user record by email for authentication
SELECT UserID, FirstName, LastName, Email, PasswordHash
FROM User
WHERE Email = 'student1@nyu.edu';

-- Direct credential check example
SELECT UserID, FirstName, LastName
FROM User
WHERE Email = 'student1@nyu.edu'
  AND PasswordHash = 'hashed_pw_1';


-- JOIN REQUESTS
```

```
-- User submits a request to join a club
INSERT INTO JoinRequest (RequestID, RequestStatus, RequestTime, UserID, ClubID)
VALUES ('R0001', 'Pending', NOW(), 'U0001', 'CL001');

-- View all pending requests for a specific club
SELECT JR.RequestID, JR.UserID, U.FirstName, U.LastName, JR.RequestStatus,
JR.RequestTime
FROM JoinRequest JR
INNER JOIN User U ON JR.UserID = U.UserID
WHERE JR.ClubID = 'CL001'
  AND JR.RequestStatus = 'Pending';

-- Approve a join request
UPDATE JoinRequest
SET RequestStatus = 'Approved'
WHERE RequestID = 'R0001';

-- Reject a join request
UPDATE JoinRequest
SET RequestStatus = 'Rejected'
WHERE RequestID = 'R0002';

-- View all join requests submitted by a specific user
SELECT *
FROM JoinRequest
WHERE UserID = 'U0001';

-- Check whether a user already has a pending request for a club
SELECT *
FROM JoinRequest
WHERE UserID = 'U0001'
  AND ClubID = 'CL001'
  AND RequestStatus = 'Pending';


-- CLUB MEMBERSHIP MANAGEMENT

-- Add a user as a member of a club
```

```
INSERT INTO ClubMembership (UserID, ClubID, MembershipRole, MembershipStatus,
MembershipJoinDate)
VALUES ('U0001', 'CL001', 'Member', 'Active', CURDATE());

-- View all members of a specific club
SELECT CM.UserID, U.FirstName, U.LastName, CM.MembershipRole,
CM.MembershipStatus, CM.MembershipJoinDate
FROM ClubMembership CM
INNER JOIN User U ON CM.UserID = U.UserID
WHERE CM.ClubID = 'CL001';

-- View all clubs that a specific user belongs to
SELECT C.ClubID, C.ClubName, CM.MembershipRole, CM.MembershipStatus
FROM ClubMembership CM
INNER JOIN Club C ON CM.ClubID = C.ClubID
WHERE CM.UserID = 'U0001';

-- Update a member's role
UPDATE ClubMembership
SET MembershipRole = 'Officer'
WHERE UserID = 'U0001' AND ClubID = 'CL001';

-- Update a member's status
UPDATE ClubMembership
SET MembershipStatus = 'Inactive'
WHERE UserID = 'U0001' AND ClubID = 'CL001';

-- Remove a user from a club
DELETE FROM ClubMembership
WHERE UserID = 'U0001' AND ClubID = 'CL001';


-- STORED PROCEDURE FOR APPROVING A JOIN REQUEST

DELIMITER $$

CREATE PROCEDURE ApproveJoinRequest(IN p_RequestID CHAR(5))
BEGIN
   DECLARE v_UserID CHAR(5);
   DECLARE v_ClubID CHAR(5);
```

```
    -- Get the user and club from the join request
    SELECT UserID, ClubID
    INTO v_UserID, v_ClubID
    FROM JoinRequest
    WHERE RequestID = p_RequestID;

    -- Update the join request status to Approved
    UPDATE JoinRequest
    SET RequestStatus = 'Approved'
    WHERE RequestID = p_RequestID;

    -- Add the user to the club membership table only if not already a member
    IF NOT EXISTS (
        SELECT 1
        FROM ClubMembership
        WHERE UserID = v_UserID
          AND ClubID = v_ClubID
    ) THEN
        INSERT INTO ClubMembership (UserID, ClubID, MembershipRole, MembershipStatus,
MembershipJoinDate)
        VALUES (v_UserID, v_ClubID, 'Member', 'Active', CURDATE());
    END IF;
END$$

DELIMITER ;


-- TRIGGER TO PREVENT DUPLICATE PENDING JOIN REQUESTS

DELIMITER $$

CREATE TRIGGER PreventDuplicatePendingJoinRequest
BEFORE INSERT ON JoinRequest
FOR EACH ROW
BEGIN
    IF EXISTS (
        SELECT 1
        FROM JoinRequest
        WHERE UserID = NEW.UserID
```

```
        AND ClubID = NEW.ClubID
        AND RequestStatus = 'Pending'
    ) THEN
      SIGNAL SQLSTATE '45000'
      SET MESSAGE_TEXT = 'User already has a pending join request for this club';
    END IF;
END$$

DELIMITER ;
```

Events Sql:

```
USE club_organizations;

-- INSERT LOCATIONS
INSERT INTO Location (LocationID, BuildingName, RoomNumber, HomeAddress)
VALUES ('LO001', 'Rogers Hall', 101, '6 MetroTech Center, Brooklyn, NY 11201');
INSERT INTO Location (LocationID, BuildingName, RoomNumber, HomeAddress)
VALUES ('LO002', 'Jacobs Academic Building', 674, '370 Jay Street, Brooklyn, NY
11201');
INSERT INTO Location (LocationID, BuildingName, RoomNumber, HomeAddress)
VALUES ('LO003', 'Dibner Library', 201, '5 MetroTech Center, Brooklyn, NY 11201');
INSERT INTO Location (LocationID, BuildingName, RoomNumber, HomeAddress)
VALUES ('LO004', 'Tandon MakerSpace', 100, '6 MetroTech Center, Brooklyn, NY
11201');
INSERT INTO Location (LocationID, BuildingName, RoomNumber, HomeAddress)
VALUES ('LO005', 'Pfizer Auditorium', 1, '5 MetroTech Center, Brooklyn, NY 11201');
INSERT INTO Location (LocationID, BuildingName, RoomNumber, HomeAddress)
VALUES ('LO006', '2 MetroTech Center', 802, '2 MetroTech Center, Brooklyn, NY
11201');
INSERT INTO Location (LocationID, BuildingName, RoomNumber, HomeAddress)
VALUES ('LO007', 'Rogers Hall', 310, '6 MetroTech Center, Brooklyn, NY 11201');
INSERT INTO Location (LocationID, BuildingName, RoomNumber, HomeAddress)
VALUES ('LO008', 'Jacobs Academic Building', 473, '370 Jay Street, Brooklyn, NY
11201');
INSERT INTO Location (LocationID, BuildingName, RoomNumber, HomeAddress)
VALUES ('LO009', 'Dibner Library', 105, '5 MetroTech Center, Brooklyn, NY 11201');
```

```
INSERT INTO Location (LocationID, BuildingName, RoomNumber, HomeAddress)
VALUES ('LO010', '2 MetroTech Center', 1001, '2 MetroTech Center, Brooklyn, NY
11201');

-- INSERT EVENT TYPES
INSERT INTO EventType (EventTypeID, EventTypeName) VALUES ('ET001',
'Workshop');
INSERT INTO EventType (EventTypeID, EventTypeName) VALUES ('ET002',
'Networking');
INSERT INTO EventType (EventTypeID, EventTypeName) VALUES ('ET003', 'Social');
INSERT INTO EventType (EventTypeID, EventTypeName) VALUES ('ET004', 'General
Meeting');
INSERT INTO EventType (EventTypeID, EventTypeName) VALUES ('ET005',
'Hackathon');
INSERT INTO EventType (EventTypeID, EventTypeName) VALUES ('ET006',
'Seminar');
INSERT INTO EventType (EventTypeID, EventTypeName) VALUES ('ET007',
'Competition');
INSERT INTO EventType (EventTypeID, EventTypeName) VALUES ('ET008',
'Community Service');
INSERT INTO EventType (EventTypeID, EventTypeName) VALUES ('ET009', 'Info
Session');
INSERT INTO EventType (EventTypeID, EventTypeName) VALUES ('ET010',
'Conference');

-- INSERT EVENTS
INSERT INTO Event (EventID, EventTitle, EventDescription, EventStartTime,
EventEndTime, EventCapacity, EventStatus, EventVisibility, EventTypeID, LocationID,
ClubID) VALUES
('EV001', 'SASE Networking Night', 'Connect with industry professionals and SASE
alumni over food and conversation.', '2025-10-15 18:00:00', '2025-10-15 20:00:00', 80,
'Scheduled', 'Public', 'ET002', 'LO005', 'CL001');

INSERT INTO Event (EventID, EventTitle, EventDescription, EventStartTime,
EventEndTime, EventCapacity, EventStatus, EventVisibility, EventTypeID, LocationID,
ClubID) VALUES
('EV002', 'AOE Sisters Social', 'A members-only social for current AOE sisters.',
'2025-10-20 17:00:00', '2025-10-20 19:00:00', 30, 'Scheduled', 'MembersOnly', 'ET003',
'LO002', 'CL002');
```

INSERT INTO Event (EventID, EventTitle, EventDescription, EventStartTime, EventEndTime, EventCapacity, EventStatus, EventVisibility, EventTypeID, LocationID, ClubID) VALUES
('EV003', 'BFG Stock Pitch Competition', 'Compete to deliver the best stock pitch in front of industry judges.', '2025-11-05 14:00:00', '2025-11-05 17:00:00', 50, 'Scheduled', 'Public', 'ET007', 'LO005', 'CL003');

INSERT INTO Event (EventID, EventTitle, EventDescription, EventStartTime, EventEndTime, EventCapacity, EventStatus, EventVisibility, EventTypeID, LocationID, ClubID) VALUES
('EV004', 'Chem Club Lab Tour', 'Tour the chemistry research labs with faculty guides. Open to members only.', '2025-09-28 11:00:00', '2025-09-28 12:30:00', 20, 'Completed', 'MembersOnly', 'ET009', 'LO007', 'CL004');

INSERT INTO Event (EventID, EventTitle, EventDescription, EventStartTime, EventEndTime, EventCapacity, EventStatus, EventVisibility, EventTypeID, LocationID, ClubID) VALUES
('EV005', 'oSTEM Study Session', 'Midterm study session open to all students. Snacks provided.', '2025-10-08 16:00:00', '2025-10-08 19:00:00', 40, 'Completed', 'Public', 'ET003', 'LO003', 'CL005');

INSERT INTO Event (EventID, EventTitle, EventDescription, EventStartTime, EventEndTime, EventCapacity, EventStatus, EventVisibility, EventTypeID, LocationID, ClubID) VALUES
('EV006', 'Anime Movie Night', 'Watch a classic Miyazaki film with club members. Popcorn and drinks included.', '2025-11-12 19:00:00', '2025-11-12 22:00:00', 60, 'Scheduled', 'Public', 'ET003', 'LO005', 'CL006');

INSERT INTO Event (EventID, EventTitle, EventDescription, EventStartTime, EventEndTime, EventCapacity, EventStatus, EventVisibility, EventTypeID, LocationID, ClubID) VALUES
('EV007', 'PPC Weekly Practice', 'Weekly competitive programming practice session. All skill levels welcome.', '2025-10-22 18:00:00', '2025-10-22 20:00:00', 35, 'Scheduled', 'MembersOnly', 'ET004', 'LO008', 'CL007');

INSERT INTO Event (EventID, EventTitle, EventDescription, EventStartTime, EventEndTime, EventCapacity, EventStatus, EventVisibility, EventTypeID, LocationID, ClubID) VALUES

('EV008', 'Robotics Arduino Workshop', 'Hands-on Arduino workshop. Build your first robot arm.', '2025-11-01 13:00:00', '2025-11-01 16:00:00', 25, 'Scheduled', 'Public', 'ET001', 'LO004', 'CL008');

INSERT INTO Event (EventID, EventTitle, EventDescription, EventStartTime, EventEndTime, EventCapacity, EventStatus, EventVisibility, EventTypeID, LocationID, ClubID) VALUES
('EV009', 'SWE Resume Workshop', 'Get your resume reviewed by engineers from top companies.', '2025-09-18 15:00:00', '2025-09-18 17:00:00', 45, 'Completed', 'Public', 'ET001', 'LO002', 'CL009');

INSERT INTO Event (EventID, EventTitle, EventDescription, EventStartTime, EventEndTime, EventCapacity, EventStatus, EventVisibility, EventTypeID, LocationID, ClubID) VALUES
('EV010', 'Student Council Town Hall', 'Open forum for students to voice concerns and ideas for the semester.', '2025-10-01 12:00:00', '2025-10-01 13:30:00', 100, 'Completed', 'Public', 'ET004', 'LO005', 'CL010');

INSERT INTO Event (EventID, EventTitle, EventDescription, EventStartTime, EventEndTime, EventCapacity, EventStatus, EventVisibility, EventTypeID, LocationID, ClubID) VALUES
('EV011', 'SASE Hackathon 2025', 'A 24-hour hackathon hosted by SASE. Teams of up to 4. Prizes for top 3.', '2025-11-15 10:00:00', '2025-11-16 10:00:00', 120, 'Scheduled', 'Public', 'ET005', 'LO006', 'CL001');

INSERT INTO Event (EventID, EventTitle, EventDescription, EventStartTime, EventEndTime, EventCapacity, EventStatus, EventVisibility, EventTypeID, LocationID, ClubID) VALUES
('EV012', 'SWE Industry Panel', 'Hear from women engineers at Meta, Google, and Amazon about their career paths.', '2025-11-20 17:00:00', '2025-11-20 19:00:00', 70, 'Scheduled', 'Public', 'ET006', 'LO005', 'CL009');

-- INSERT RSVPs
INSERT INTO RSVP (RSVPID, RSVPStatus, RSVPCreationDate, UserID, EventID) VALUES ('RS001', 'Going', '2025-10-10', 'U0001', 'EV001');
INSERT INTO RSVP (RSVPID, RSVPStatus, RSVPCreationDate, UserID, EventID) VALUES ('RS002', 'Going', '2025-10-11', 'U0002', 'EV001');
INSERT INTO RSVP (RSVPID, RSVPStatus, RSVPCreationDate, UserID, EventID) VALUES ('RS003', 'Tentative', '2025-10-12', 'U0003', 'EV001');

```
INSERT INTO RSVP (RSVPID, RSVPStatus, RSVPCreationDate, UserID, EventID)
VALUES ('RS004', 'Going', '2025-10-18', 'U0002', 'EV002');
INSERT INTO RSVP (RSVPID, RSVPStatus, RSVPCreationDate, UserID, EventID)
VALUES ('RS005', 'Going', '2025-10-30', 'U0004', 'EV003');
INSERT INTO RSVP (RSVPID, RSVPStatus, RSVPCreationDate, UserID, EventID)
VALUES ('RS006', 'Going', '2025-09-25', 'U0003', 'EV004');
INSERT INTO RSVP (RSVPID, RSVPStatus, RSVPCreationDate, UserID, EventID)
VALUES ('RS007', 'Going', '2025-10-05', 'U0005', 'EV005');
INSERT INTO RSVP (RSVPID, RSVPStatus, RSVPCreationDate, UserID, EventID)
VALUES ('RS008', 'NotGoing', '2025-10-06', 'U0006', 'EV005');
INSERT INTO RSVP (RSVPID, RSVPStatus, RSVPCreationDate, UserID, EventID)
VALUES ('RS009', 'Going', '2025-11-10', 'U0007', 'EV006');
INSERT INTO RSVP (RSVPID, RSVPStatus, RSVPCreationDate, UserID, EventID)
VALUES ('RS010', 'Going', '2025-10-20', 'U0008', 'EV007');
INSERT INTO RSVP (RSVPID, RSVPStatus, RSVPCreationDate, UserID, EventID)
VALUES ('RS011', 'Tentative', '2025-10-28', 'U0009', 'EV008');
INSERT INTO RSVP (RSVPID, RSVPStatus, RSVPCreationDate, UserID, EventID)
VALUES ('RS012', 'Going', '2025-09-15', 'U0010', 'EV009');

-- INSERT ATTENDANCE
INSERT INTO Attendance(CheckInTime, CheckInMethod, EventID, RSVPID)
VALUES ('2025-10-15 17:55:00', 'Manual', 'EV001', 'RS001');
INSERT INTO Attendance(CheckInTime, CheckInMethod, EventID, RSVPID)
VALUES ('2025-10-15 18:02:00', 'Manual', 'EV001', 'RS002');
INSERT INTO Attendance(CheckInTime, CheckInMethod, EventID, RSVPID)
VALUES ('2025-10-20 16:58:00', 'QRCode', 'EV002', 'RS004');
INSERT INTO Attendance(CheckInTime, CheckInMethod, EventID, RSVPID)
VALUES ('2025-11-05 13:50:00', 'Manual', 'EV003', 'RS005');
INSERT INTO Attendance(CheckInTime, CheckInMethod, EventID, RSVPID)
VALUES ('2025-09-28 10:55:00', 'QRCode', 'EV004', 'RS006');
INSERT INTO Attendance(CheckInTime, CheckInMethod, EventID, RSVPID)
VALUES ('2025-10-08 15:58:00', 'Manual', 'EV005', 'RS007');
INSERT INTO Attendance(CheckInTime, CheckInMethod, EventID, RSVPID)
VALUES ('2025-11-12 18:55:00', 'QRCode', 'EV006', 'RS009');
INSERT INTO Attendance(CheckInTime, CheckInMethod, EventID, RSVPID)
VALUES ('2025-10-22 17:50:00', 'Manual', 'EV007', 'RS010');
INSERT INTO Attendance(CheckInTime, CheckInMethod, EventID, RSVPID)
VALUES ('2025-09-18 14:55:00', 'QRCode', 'EV009', 'RS012');
INSERT INTO Attendance(CheckInTime, CheckInMethod, EventID, RSVPID)
VALUES ('2025-11-01 11:50:00', 'Manual', 'EV008', 'RS011');
```

```
-- SQL COMMANDS USED BY THE APP


-- EVENT CREATION --

-- Insert a new event
INSERT INTO Event (EventID, EventTitle, EventDescription, EventStartTime,
EventEndTime, EventCapacity, EventStatus, EventVisibility, EventTypeID, LocationID,
ClubID)
VALUES ('EV013', 'New Workshop', 'A brand new workshop for beginners.',
'2025-12-01 10:00:00', '2025-12-01 12:00:00', 30, 'Scheduled', 'Public', 'ET001', 'LO001',
'CL001');

-- Update event details
UPDATE Event
SET EventTitle = 'Updated Workshop Title',
    EventDescription = 'Updated description with new details.',
    EventCapacity = 40
WHERE EventID = 'EV013';

-- Cancel an event
UPDATE Event
SET EventStatus = 'Cancelled'
WHERE EventID = 'EV013';

-- Delete an event
DELETE FROM Event
WHERE EventID = 'EV013';


-- EVENT SEARCH AND FILTERING --

-- Get all upcoming scheduled events
SELECT e.EventID, e.EventTitle, e.EventStartTime, e.EventEndTime,
       e.EventVisibility, c.ClubName, l.BuildingName, l.RoomNumber
FROM Event e
INNER JOIN Club c ON e.ClubID = c.ClubID
INNER JOIN Location l ON e.LocationID = l.LocationID
```

```
WHERE e.EventStatus = 'Scheduled'
ORDER BY e.EventStartTime;

-- Search events by title keyword
SELECT e.EventID, e.EventTitle, e.EventStartTime, c.ClubName
FROM Event e
INNER JOIN Club c ON e.ClubID = c.ClubID
WHERE e.EventTitle LIKE '%Workshop%';

-- Filter events by club
SELECT EventID, EventTitle, EventStartTime, EventEndTime, EventVisibility
FROM Event
WHERE ClubID = 'CL001'
ORDER BY EventStartTime;

-- Filter events by event type
SELECT e.EventID, e.EventTitle, e.EventStartTime, c.ClubName
FROM Event e
INNER JOIN Club c ON e.ClubID = c.ClubID
WHERE e.EventTypeID = 'ET001';

-- Filter public events only
SELECT e.EventID, e.EventTitle, e.EventStartTime, c.ClubName, l.BuildingName
FROM Event e
INNER JOIN Club c ON e.ClubID = c.ClubID
INNER JOIN Location l ON e.LocationID = l.LocationID
WHERE e.EventVisibility = 'Public' AND e.EventStatus = 'Scheduled'
ORDER BY e.EventStartTime;

-- Filter events by date range
SELECT e.EventID, e.EventTitle, e.EventStartTime, c.ClubName
FROM Event e
INNER JOIN Club c ON e.ClubID = c.ClubID
WHERE e.EventStartTime BETWEEN '2025-10-01' AND '2025-10-31';

-- Get single event details (individual event page)
SELECT e.EventID, e.EventTitle, e.EventDescription, e.EventStartTime,
e.EventEndTime,
    e.EventCapacity, e.EventVisibility, e.EventStatus,
    c.ClubName, l.BuildingName, l.RoomNumber, l.HomeAddress
```

```
FROM Event e
INNER JOIN Club c ON e.ClubID = c.ClubID
INNER JOIN Location l ON e.LocationID = l.LocationID
WHERE e.EventID = 'EV001';

-- Get events a user has RSVP'd to
SELECT e.EventID, e.EventTitle, e.EventStartTime, r.RSVPStatus, c.ClubName
FROM Event e
INNER JOIN RSVP r ON e.EventID = r.EventID
INNER JOIN Club c ON e.ClubID = c.ClubID
WHERE r.UserID = 'U0001';

-- Get RSVP count for an event
SELECT e.EventID, e.EventTitle, e.EventCapacity, COUNT(r.RSVPID) AS GoingCount
FROM Event e
LEFT JOIN RSVP r ON e.EventID = r.EventID AND r.RSVPStatus = 'Going'
WHERE e.EventID = 'EV001'
GROUP BY e.EventID, e.EventTitle, e.EventCapacity;


-- RSVP HANDLING --

-- Create a new RSVP
INSERT INTO RSVP (RSVPID, RSVPStatus, RSVPCreationDate, UserID, EventID)
VALUES ('RS013', 'Going', CURDATE(), 'U0001', 'EV006');

-- Update RSVP status
UPDATE RSVP
SET RSVPStatus = 'Tentative'
WHERE RSVPID = 'RS013';

-- Cancel RSVP
UPDATE RSVP
SET RSVPStatus = 'NotGoing'
WHERE RSVPID = 'RS013';

-- Delete an RSVP
DELETE FROM RSVP
WHERE RSVPID = 'RS013';
```

```
-- Check if user already RSVP'd to an event
SELECT RSVPID, RSVPStatus
FROM RSVP
WHERE UserID = 'U0001' AND EventID = 'EV006';

-- Get all RSVPs for an event
SELECT r.RSVPID, r.RSVPStatus, r.RSVPCreationDate, u.FirstName, u.LastName
FROM RSVP r
INNER JOIN User u ON r.UserID = u.UserID
WHERE r.EventID = 'EV001';

-- Count RSVPs by status for an event
SELECT RSVPStatus, COUNT(*) AS StatusCount
FROM RSVP
WHERE EventID = 'EV001'
GROUP BY RSVPStatus;


-- ATTENDANCE / CHECK-IN --

-- Check in a user
INSERT INTO Attendance (CheckInTime, CheckInMethod, EventID, RSVPID)
VALUES (NOW(), 'Manual', 'EV001', 'RS001');

-- Check if user already checked in
SELECT CheckInTime, CheckInMethod
FROM Attendance
WHERE EventID = 'EV001' AND RSVPID = 'RS001';

-- Get all attendees for an event
SELECT u.FirstName, u.LastName, a.CheckInTime, a.CheckInMethod
FROM Attendance a
INNER JOIN RSVP r ON a.RSVPID = r.RSVPID
INNER JOIN User u ON r.UserID = u.UserID
WHERE a.EventID = 'EV001';

-- Count attendees for an event
SELECT COUNT(*) AS TotalAttendees
FROM Attendance
WHERE EventID = 'EV001';
```

```
-- Verify user RSVP'd Going before check-in
SELECT r.RSVPID
FROM RSVP r
WHERE r.EventID = 'EV001' AND r.UserID = 'U0001' AND r.RSVPStatus = 'Going';

-- Delete attendance record
DELETE FROM Attendance
WHERE EventID = 'EV001' AND RSVPID = 'RS001';


-- ADVANCED PL/SQL COMMANDS


-- TRIGGER: prevent RSVP if event is at full capacity
DELIMITER //
CREATE TRIGGER trg_check_event_capacity
BEFORE INSERT ON RSVP
FOR EACH ROW
BEGIN
   DECLARE current_going INT;
   DECLARE max_capacity INT;

   IF NEW.RSVPStatus = 'Going' THEN
      SELECT EventCapacity INTO max_capacity
      FROM Event WHERE EventID = NEW.EventID;

      SELECT COUNT(*) INTO current_going
      FROM RSVP WHERE EventID = NEW.EventID AND RSVPStatus = 'Going';

      IF current_going >= max_capacity THEN
         SIGNAL SQLSTATE '45000'
         SET MESSAGE_TEXT = 'Cannot RSVP: event is at full capacity.';
      END IF;
   END IF;
END //
DELIMITER ;


-- STORED PROCEDURE: create an RSVP with validation
```

```
DELIMITER //
CREATE PROCEDURE sp_create_rsvp(
   IN p_RSVPID CHAR(5),
   IN p_UserID CHAR(5),
   IN p_EventID CHAR(5),
   IN p_RSVPStatus VARCHAR(10)
)
BEGIN
   DECLARE v_event_status VARCHAR(15);
   DECLARE v_event_visibility VARCHAR(10);
   DECLARE v_club_id CHAR(5);
   DECLARE v_existing INT;
   DECLARE v_is_member INT;

   SELECT EventStatus, EventVisibility, ClubID
   INTO v_event_status, v_event_visibility, v_club_id
   FROM Event WHERE EventID = p_EventID;

   IF v_event_status != 'Scheduled' THEN
      SIGNAL SQLSTATE '45000'
      SET MESSAGE_TEXT = 'Cannot RSVP: event is not scheduled.';
   END IF;

   SELECT COUNT(*) INTO v_existing
   FROM RSVP WHERE UserID = p_UserID AND EventID = p_EventID;

   IF v_existing > 0 THEN
      SIGNAL SQLSTATE '45000'
      SET MESSAGE_TEXT = 'Cannot RSVP: user already RSVP"d to this event.';
   END IF;

   IF v_event_visibility = 'MembersOnly' THEN
      SELECT COUNT(*) INTO v_is_member
      FROM ClubMembership
      WHERE UserID = p_UserID AND ClubID = v_club_id AND MembershipStatus =
'Active';

      IF v_is_member = 0 THEN
         SIGNAL SQLSTATE '45000'
```

```
            SET MESSAGE_TEXT = 'Cannot RSVP: members-only event and user is not a
member.';
        END IF;
    END IF;

    INSERT INTO RSVP (RSVPID, RSVPStatus, RSVPCreationDate, UserID, EventID)
    VALUES (p_RSVPID, p_RSVPStatus, CURDATE(), p_UserID, p_EventID);
END //
DELIMITER ;
```